

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 01



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Recap of Previous Lecture



Topic

Cyclic group



Topic

Some important properties



Topics to be Covered



✓ Topic

Representation of graph

✓ Topic

Simple graph, multi graph and pseudo graph

✓ Topic

Maximum number of edges in a simple graph

✓ Topic

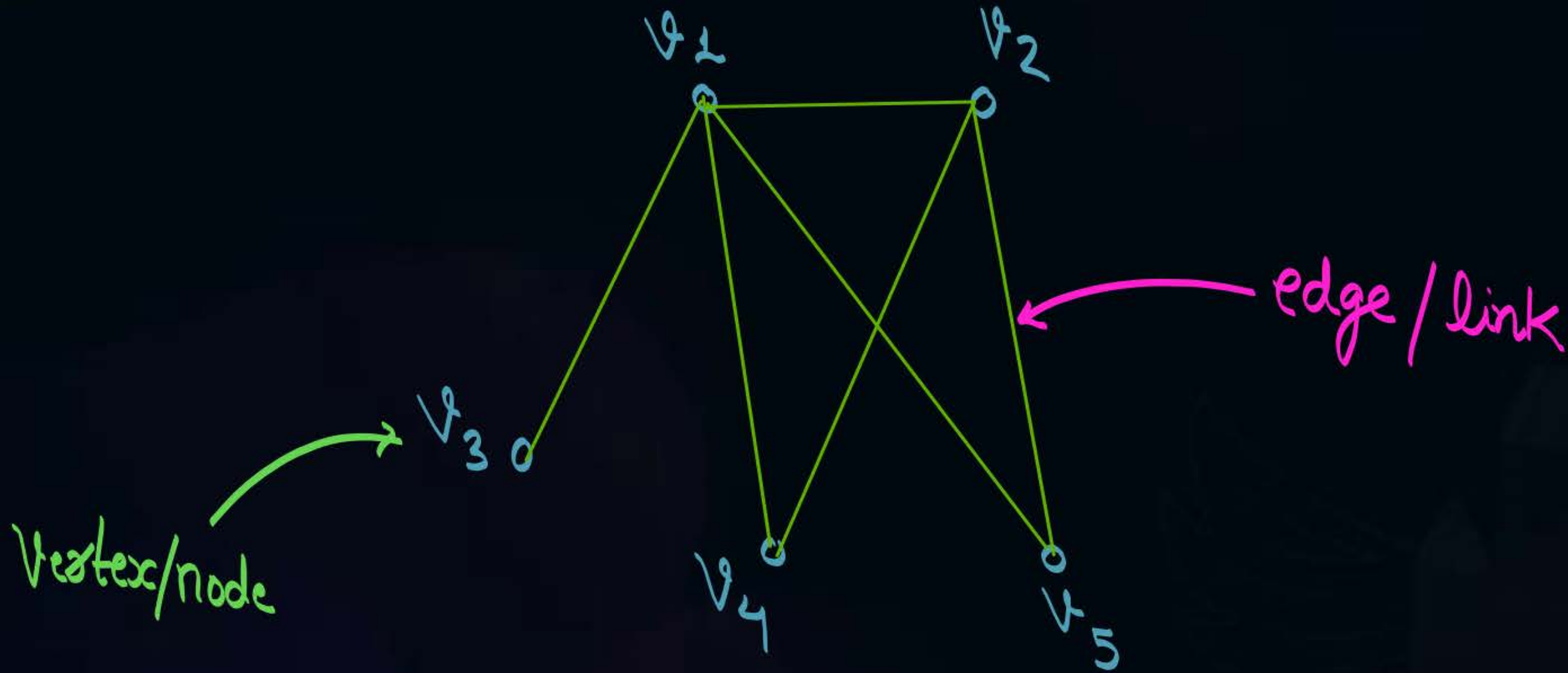
Maximum number of simple graph with n vertices

✓ Topic

Terminologies related to graphs



Topic : Graph





Topic : Graph



A Graph is defined as an **ordered pair** of two sets,

i.e., $G = (V, E)$

Name
Assigned
to the graph

Set of
Vertices

Set of edges

$|V|$ = No. of vertices (Order of graph G)

$|E|$ = No. of edges (Size of graph G)



Topic : Representation of edge

In a non-directed graph an edge is represented by a set of two vertices

$\{v_j, v_i\}$

e.g., $\{v_i, v_j\}$ which represents an edge between vertices v_i and v_j

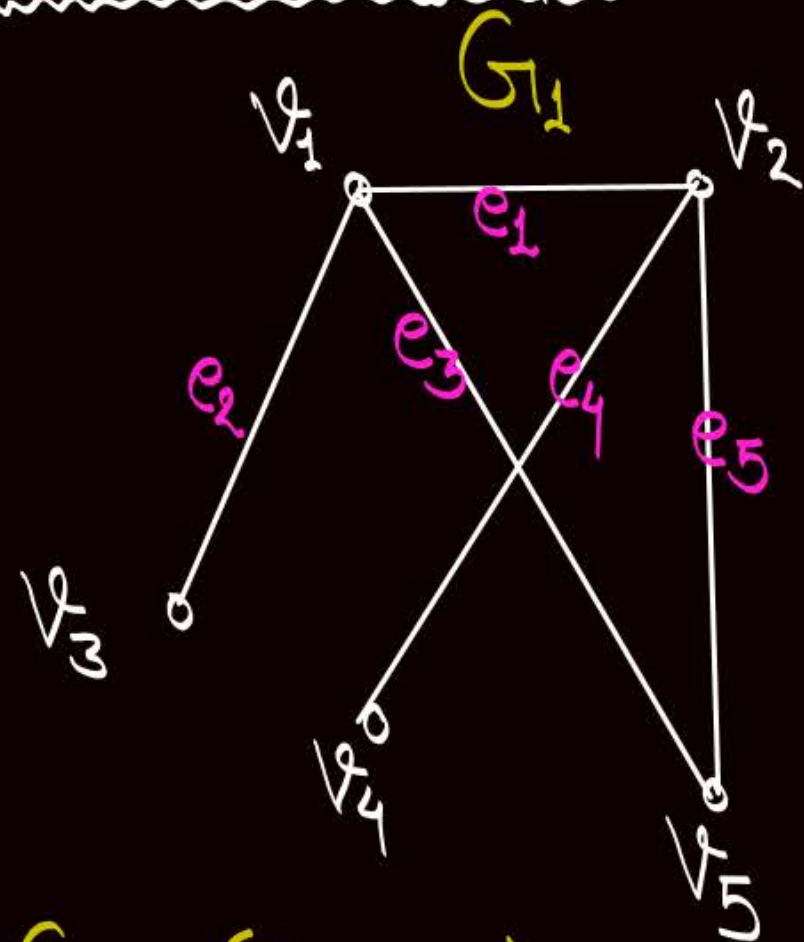
v_i & v_j are end vertices
associated with the edge

In a directed graph an edge is represented by an order pair of two vertices

e.g., (v_i, v_j) which represents an edge from vertex v_i to vertex v_j

~~(v_j, v_i)~~

④ Undirected graph



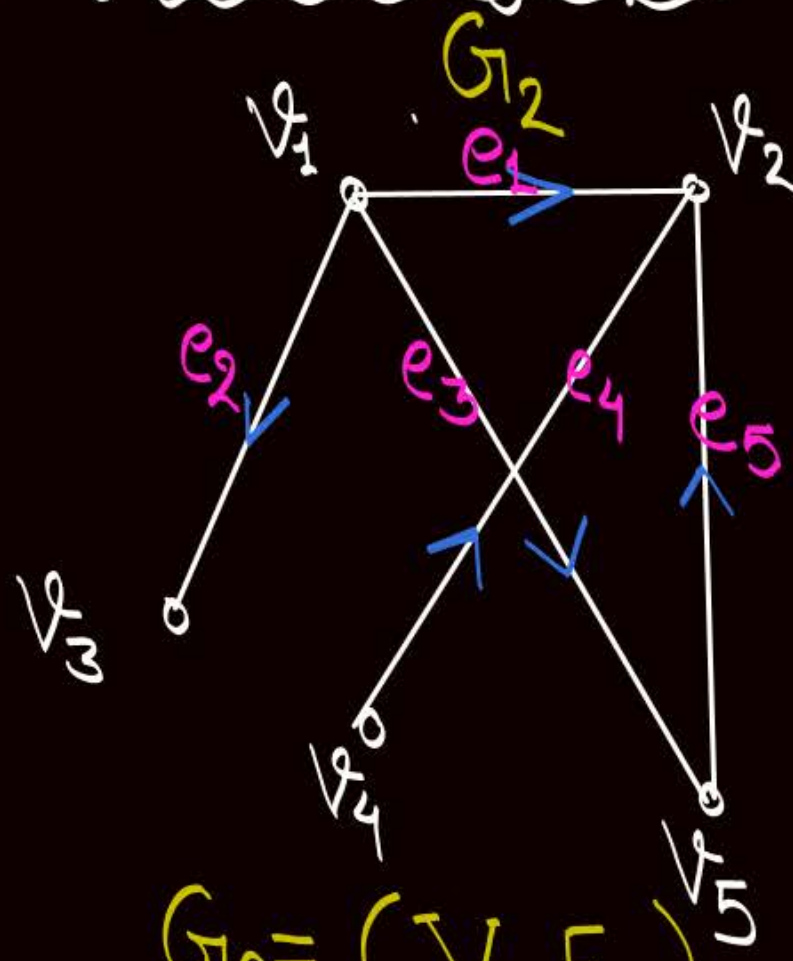
$$G_1 = (V, E)$$

$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5\}$$

$$E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}\}$$

④ Directed graph



$$G_2 = (V, E)$$

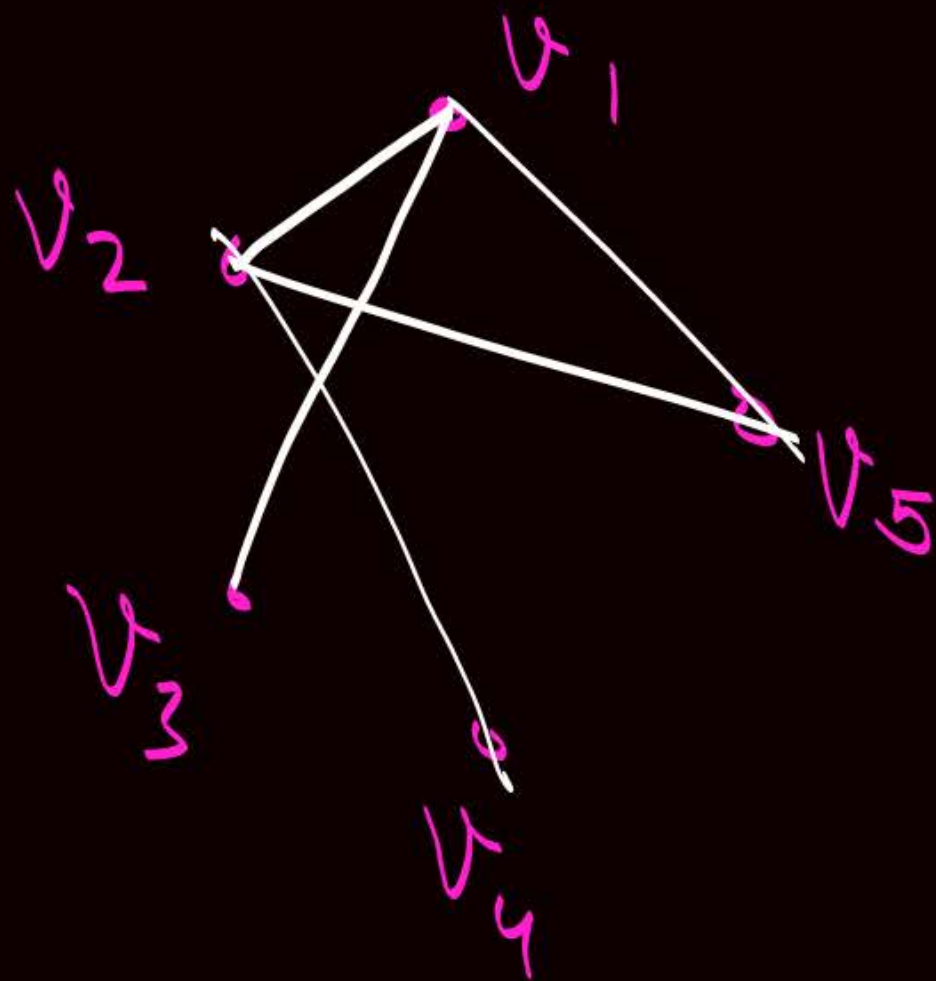
$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5\}$$

$$E = \{(v_1, v_2), (v_1, v_3), (v_1, v_5), (v_2, v_4), (v_2, v_5)\}$$

✗ $G_1 = (\{v_1, v_2, v_3, v_4, v_5\}, \{e_1, e_2, e_3, e_4, e_5\})$

✓ $G_1 = (\{v_1, v_2, v_3, v_4, v_5\}, \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}\})$

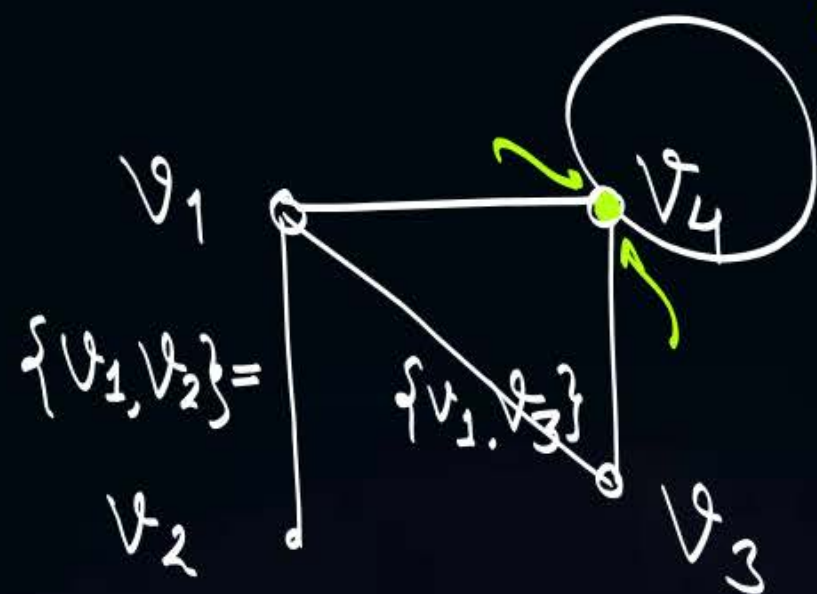




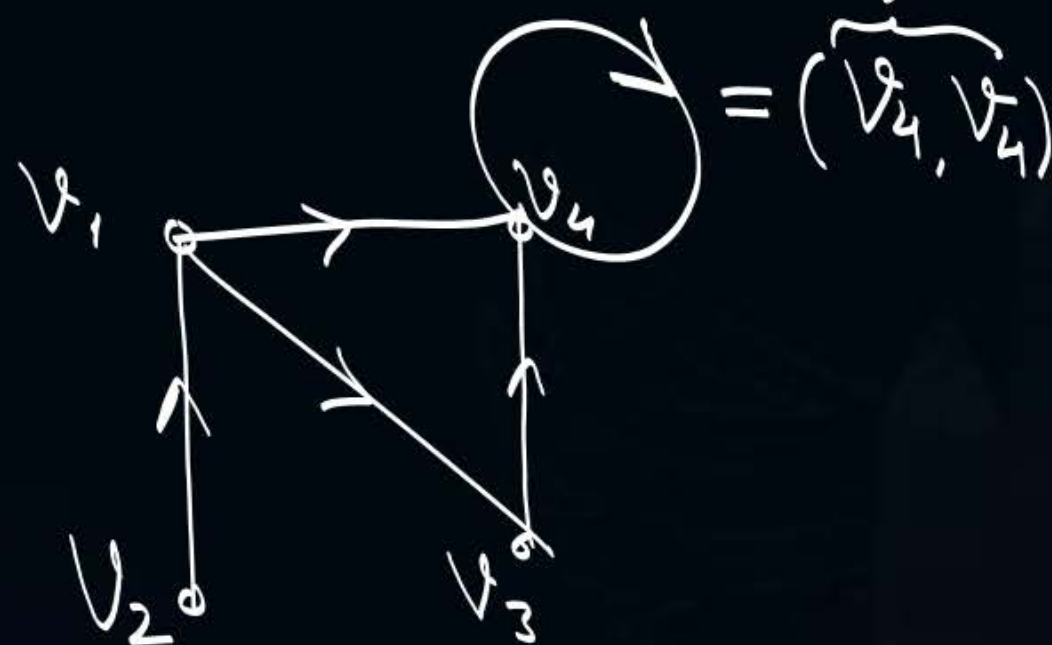
Topic : Loop



Self loop



$\{v_4, v_4\}$: Loop is an edge in which both "end vertices" are same





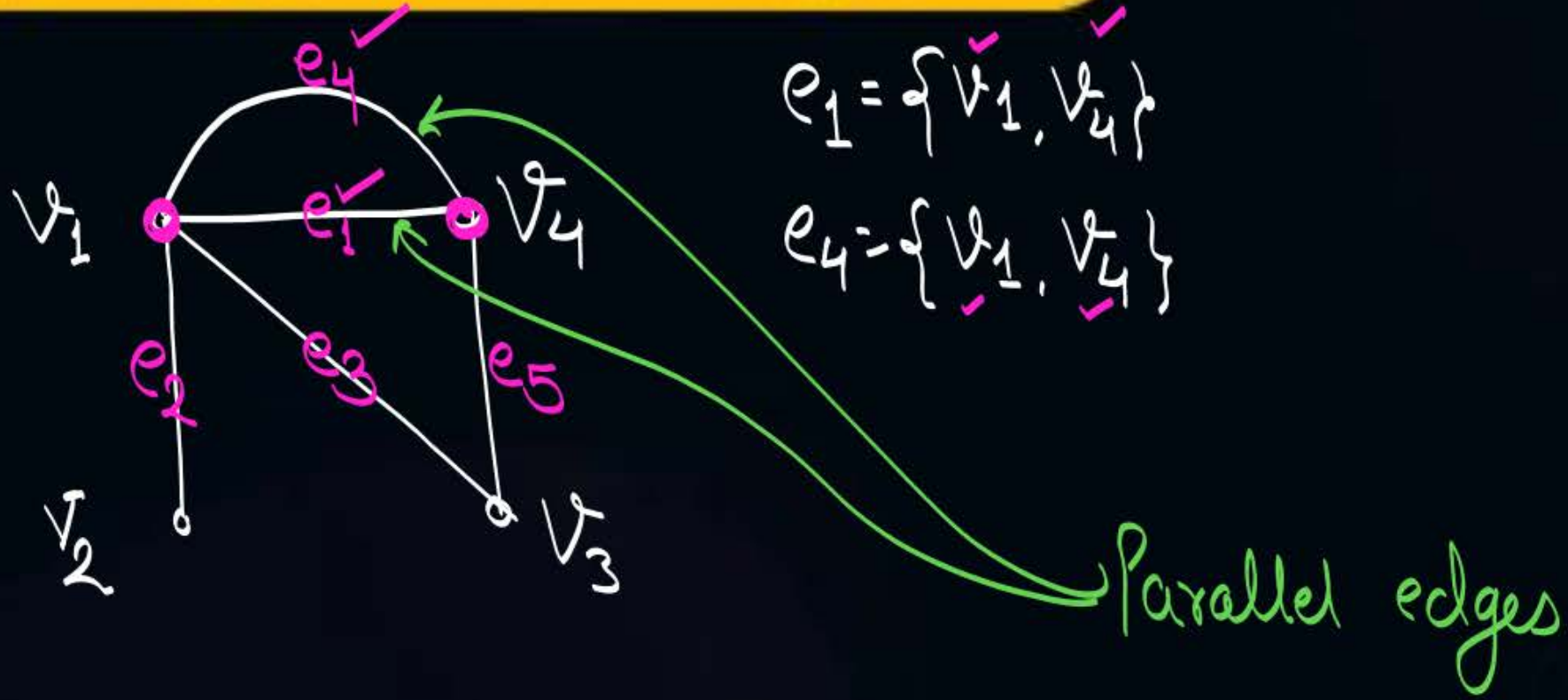
Topic : Parallel Edges



If two or more edge are associated with same end vertices, then edges are called parallel edges and resulting graph is called multi-graph.



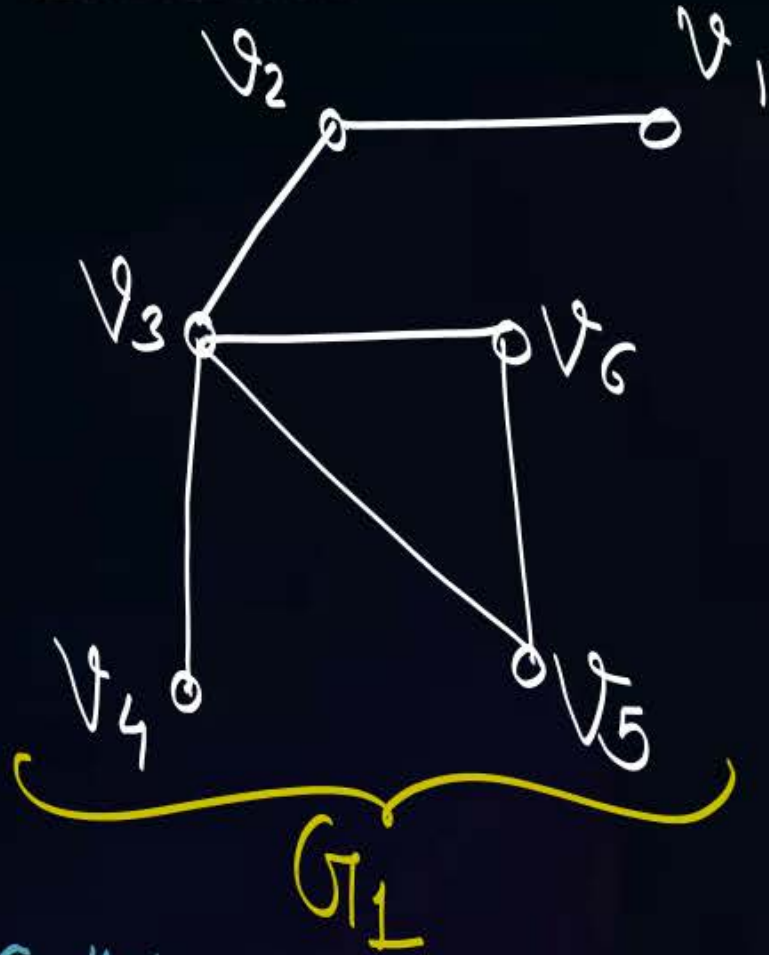
Topic : Parallel Edges



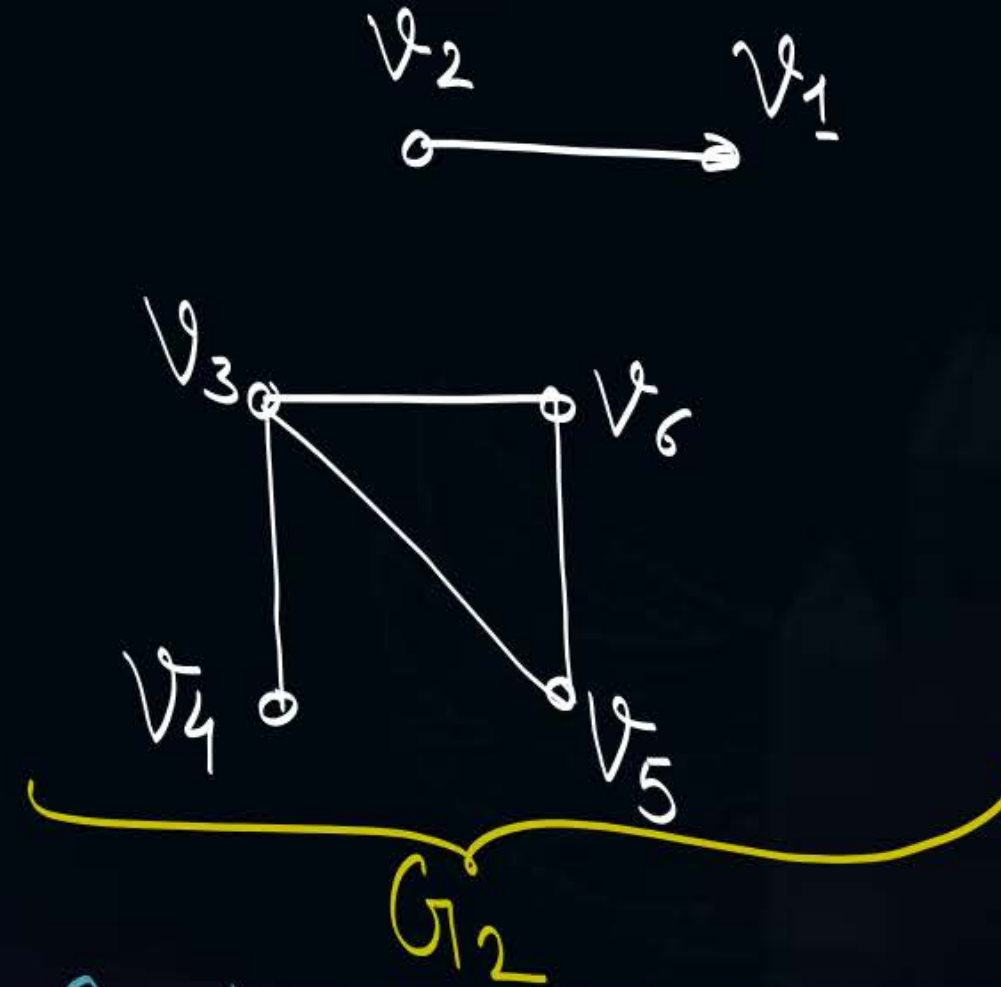


Topic : Simple graph

A graph with no self-loop and no parallel edge is called simple graph.



" G_1 " is a simple graph as well as a connected graph



G_2 is a simple graph but G_2 is a disconnected graph



Topic : Classification of graph

	Loop	Parallel edge
✓ Simple Graph	Not allowed ✗	Not allowed ✗
Multi-graph	Not allowed ✗	Allowed ✓
Pseudo graph	Allowed ✓	Allowed ✓



Topic : Simple Graph

Maximum number of edges possible in “Simple Graph” With n -vertices



Topic : Simple Graph

Maximum number of edges possible in "Simple Graph" With n-vertices

In a simple graph two distinct vertices are required to define an edge

Maximum number of
edges possible in a
simple graph with
n-vertices

No. of ways in
which two distinct
vertices can be
chosen out of
'n' vertices

$$= {}^nC_2 = \frac{n(n-1)}{2} = \frac{n^2-n}{2}$$



Topic : Simple Graph



Total Number of "Simple Graphs" possible With 'n' labeled vertices





Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

Each simple graph with 'n' vertices will contain all 'n' vertices and at most $\left(\frac{n^2-n}{2}\right)$ edges
{all simple}

ie,
No. of simple graphs possible with n-vertices

$$= \binom{\text{Choose all 'n' out of 'n' vertices}}{n} \text{ and } \binom{\text{Choose 0 or more edges out of } \left(\frac{n^2-n}{2}\right) \text{ simple edges}}{\left(\frac{n^2-n}{2}\right)}$$
$$= {}^nC_n * \left({}^{\frac{n^2-n}{2}}C_0 + {}^{\frac{n^2-n}{2}}C_1 + {}^{\frac{n^2-n}{2}}C_2 + \dots + {}^{\frac{n^2-n}{2}}C_{\frac{n^2-n}{2}} \right)$$

$\downarrow$
1

$$\therefore {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

$$\therefore \frac{n^2-n}{2} C_0 + \frac{n^2-n}{2} C_1 + \frac{n^2-n}{2} C_2 + \dots + \frac{n^2-n}{2} C_{\frac{n^2-n}{2}} = 2^{\left(\frac{n^2-n}{2}\right)} = 2^{(n_2)}$$



Topic : Simple Graph



Number of "Simple Graphs" possible With 'n' labeled vertices

$$= \left(\binom{\frac{n^2-n}{2}}{0} C_0 + \binom{\frac{n^2-n}{2}}{1} C_1 + \binom{\frac{n^2-n}{2}}{2} C_2 + \dots + \binom{\frac{n^2-n}{2}}{\frac{n^2-n}{2}} C_{\frac{n^2-n}{2}} \right)$$

$$= 2^{\binom{n^2-n}{2}} = 2^{\frac{n(n-1)}{2}}$$



Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

$$= \binom{\frac{n^2-n}{2}}{0} + \binom{\frac{n^2-n}{2}}{1} + \binom{\frac{n^2-n}{2}}{2} + \dots + \binom{\frac{n^2-n}{2}}{k} + \dots + \binom{\frac{n^2-n}{2}}{\frac{n^2-n}{2}} = 2^{\frac{n^2-n}{2}}$$

No. of simple graphs possible with n-vertices and at most 'k' edges

No. of simple graphs possible with n-vertices and exactly 'k' edges

No. of simple graphs possible with 'n' vertices and at least 'k' edges

We know $n_{C\gamma} = n_{Cn-\gamma}$

We know $n_{C\gamma} = n_{Cn-\gamma}$

In the sequenc,

$$\therefore \frac{n^2-n}{2} C_0 + \frac{n^2-n}{2} C_1 + \frac{n^2-n}{2} C_2 + \dots + \frac{n^2-n}{2} C_{\left(\frac{n^2-n}{2}-1\right)} + \frac{n^2-n}{2} C_{\frac{n^2-n}{2}}$$

$$\underbrace{\text{Summation of 1}^{st} (k+1) \text{ terms}}_{\text{III}} = \underbrace{\text{Summation of last } (k+1) \text{ terms}}$$

No. of Simple graphs
With 'n' vertices and
at most 'k' edges

= No. of Simple graphs
With 'n' vertices and
at least $\left(\frac{n^2-n}{2} - k\right)$ edges

Q:- How many simple graphs are possible with 5 labeled vertices.

$$2^{5C_2} = 2^{10} = 1024$$

Q: How many simple graphs are possible with '4' labeled vertices and at most '2' edges?

→ With '4' vertices, maximum ${}^4C_2 = 6$ edges are possible

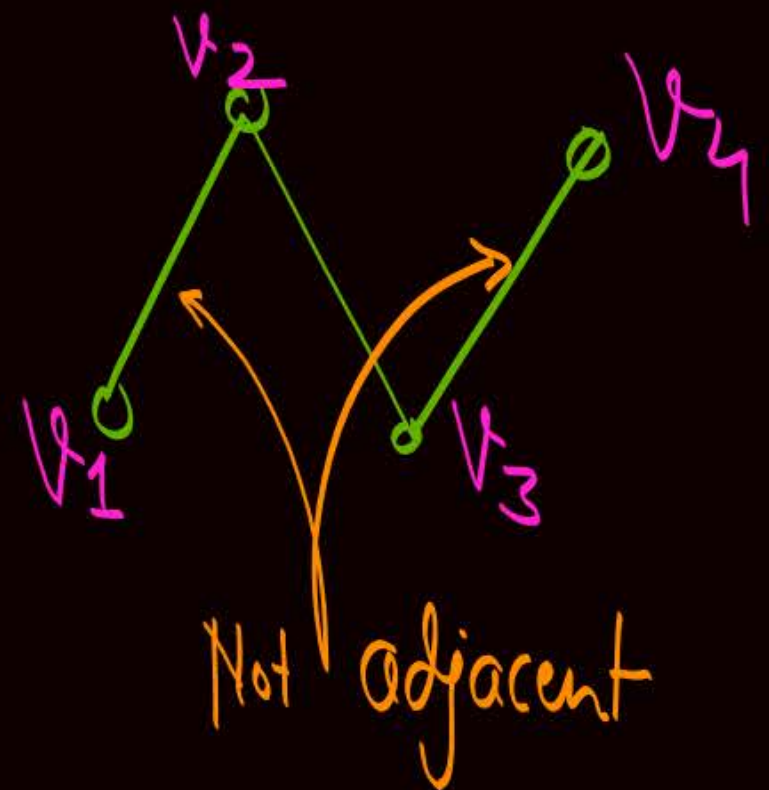
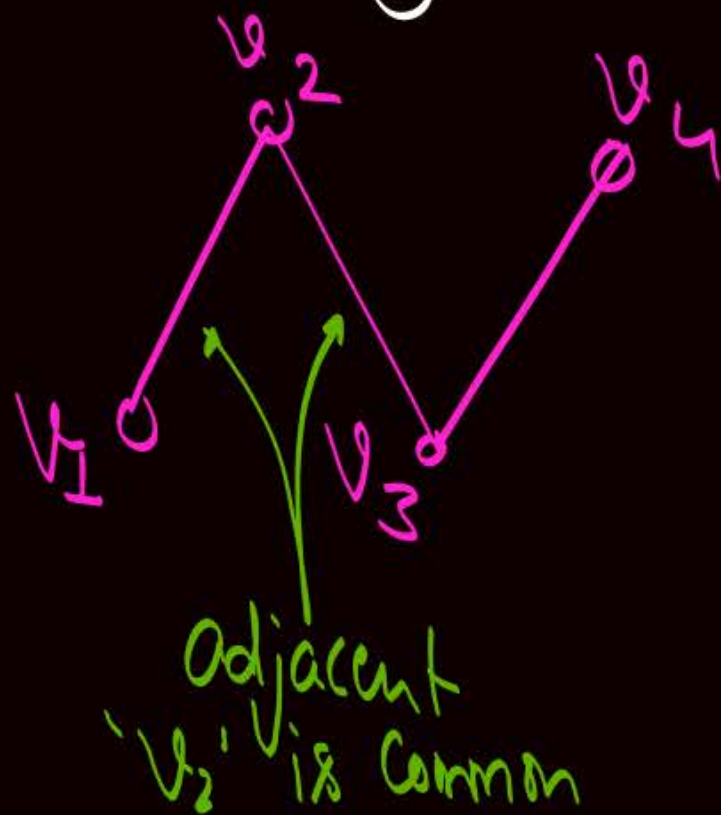
$$\therefore \text{No. of simple graphs with at most '2' edges} = {}^6C_0 + {}^6C_1 + {}^6C_2$$

$$= 1 + 6 + 15$$

$$= \boxed{22} \text{ Ans}$$

→ Adjacent vertices :— Two vertices are said to be adjacent if there exist a direct edge between them

→ Adjacent edges :— Two edges are said to be adjacent if they share a common vertex.

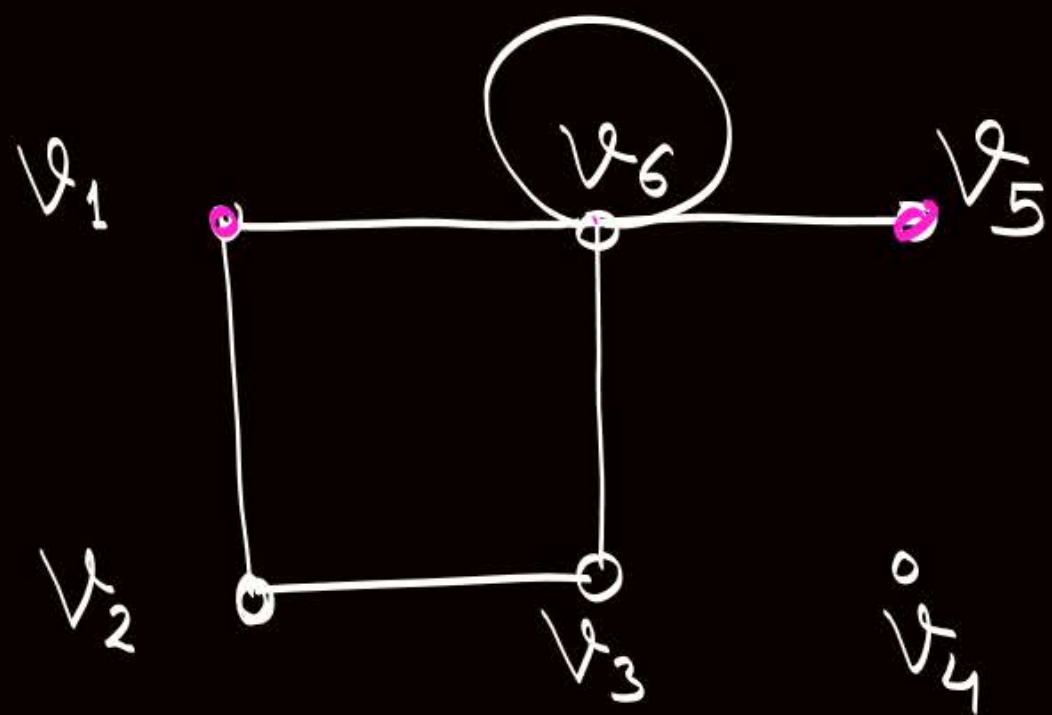




Topic : Degree of a vertex in an Undirected Graph

In an undirected graph degree of a vertex ' v ' can be denoted by $\deg(v)$ and it is defined as no. of edges incident with vertex ' v '

$$\deg(v) = \text{No. of edges incident with vertex } 'v'$$



In a non-directed graph a loop at a vertex will be counted as two edges in order to obtain the degree of that vertex

$$\deg(v_1) = 2$$

$$\deg(v_2) = 2$$

$$\deg(v_3) = 2$$

$$\deg(v_4) = 0$$

$$\deg(v_5) = 1$$

$$\deg(v_6) = 3 + 2 = 5$$



Topic : Degree of a vertex in a directed Graph

• In a directed graph we define two types of degrees for a vertex

- ① In-degree ✓
- ② Out degree ✓



Topic : Degree of a vertex in a directed Graph

In-degree :- In-degree of a vertex ' v ' is denoted by $\deg^+(v)$ and it is defined as

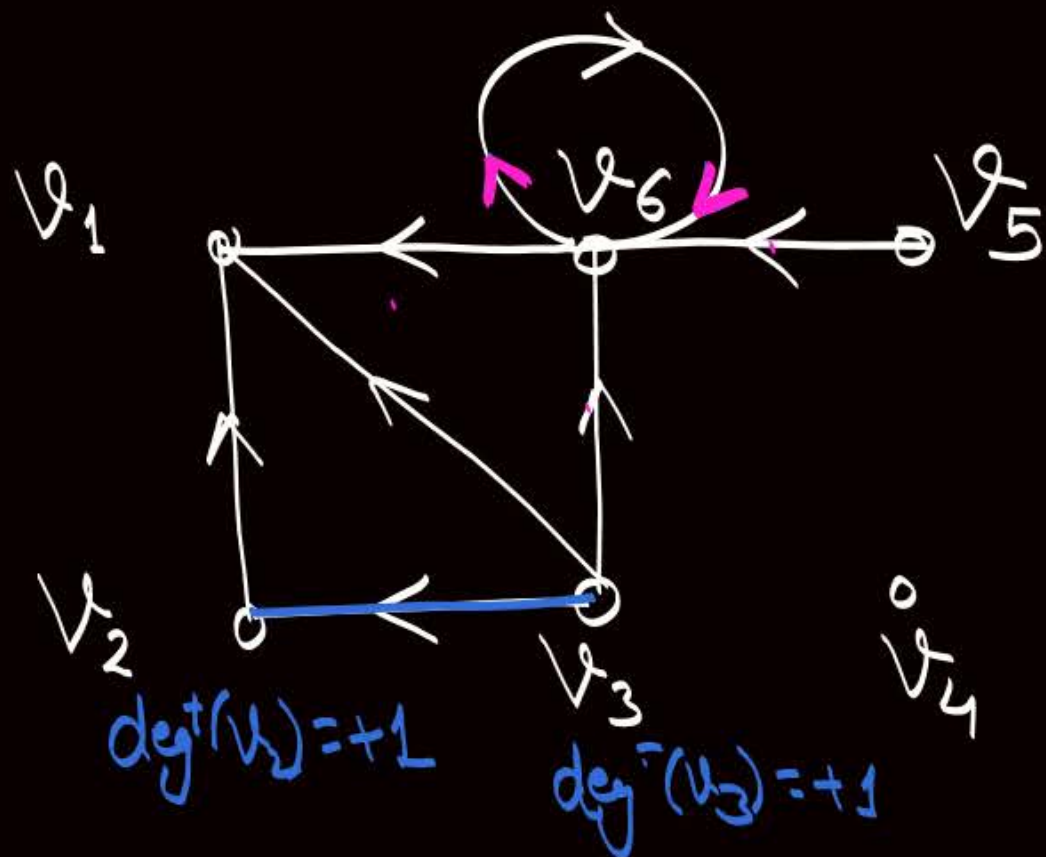
$\deg^+(v)$ = No. of edges incoming to vertex v

Out-degree :- Out-degree of a vertex ' v ' is denoted by $\deg^-(v)$, and it is defined as

$\deg^-(v)$ = No. of edges out going from vertex v

In a directed graph,
Overall degree of a vertex 'v' is

$$\deg(v) = \deg^+(v) + \deg^-(v)$$



In a directed graph a loop at a vertex will be counted as '1' edge to obtain the in-degree of that vertex and it will be counted as '1' edge to obtain the out-degree of that vertex

$$\begin{array}{l} \deg^+(v_1) = 3 \\ \deg^-(v_1) = 0 \\ \hline \Sigma = 3 \end{array} \quad \begin{array}{l} \deg^+(v_2) = 1 \\ \deg^-(v_2) = 1 \\ \hline \Sigma = 2 \end{array} \quad \begin{array}{l} \deg^+(v_3) = 0 \\ \deg^-(v_3) = 3 \\ \hline \Sigma = 3 \end{array}$$

$$\begin{array}{l} \deg^+(v_4) = 0 \\ \deg^-(v_4) = 0 \\ \hline \Sigma = 0 \end{array}$$

$$\begin{array}{l} \deg^+(v_5) = 0 \\ \deg^-(v_5) = 1 \\ \hline \Sigma = 1 \end{array}$$

$$\begin{array}{l} \deg^+(v_6) = 2 + 1 = 3 \\ \deg^-(v_6) = 1 + 1 = 2 \\ \hline \Sigma = 5 \end{array}$$

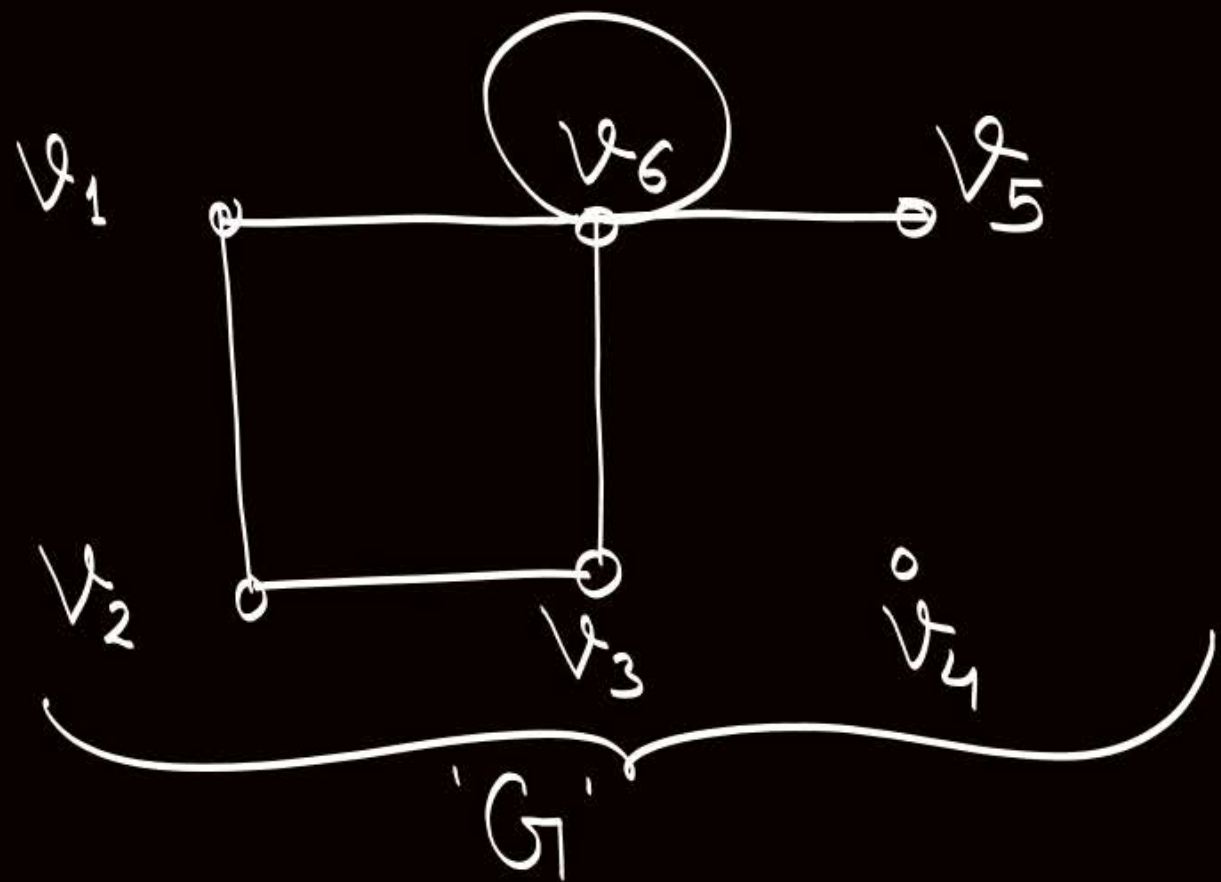


Topic : $\delta(G)$ and $\Delta(G)$

• In a graph G

$\delta(G)$: Minimum of the degrees of all the vertices of graph G is denoted by $\delta(G)$

$\Delta(G) =$ Maximum of the degrees of all the vertices of graph G is denoted by $\Delta(G)$



w.r.t. given graph,

$$\delta(G) = 0$$

$$\Delta(G) = 5$$

$$\deg(v_1) = 2$$

$$\deg(v_2) = 2$$

$$\deg(v_3) = 2$$

$$\deg(v_4) = 0$$

$$\deg(v_5) = 1$$

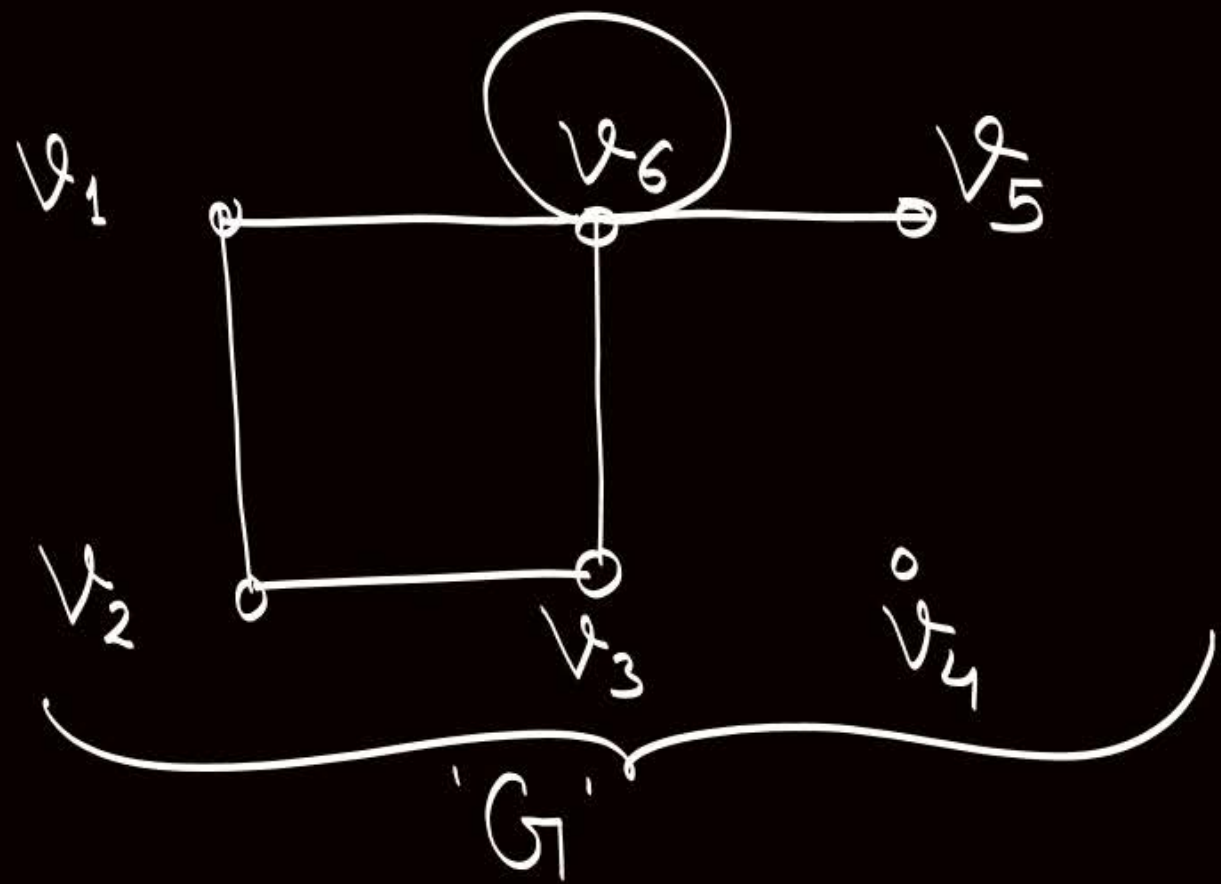
$$\deg(v_6) = 3 + 2 = 5$$



Topic : Isolated Vertex

→ In a graph G a vertex of degree $= 0$ is called an isolated vertex

i.e., if $\deg(v) = 0$, then ' v ' is an isolated vertex



$$\deg(v_1) = 2$$

$$\deg(v_2) = 2$$

$$\deg(v_3) = 2$$

$$\deg(v_4) = 0$$

$$\deg(v_5) = 1$$

$$\deg(v_6) = 3 + 2 = 5$$

' v_4 ' is an isolated vertex



Topic : Null Graph



★ A graph with no edges in it is called a NULL graph
{ vertices may be present in a NULL graph }

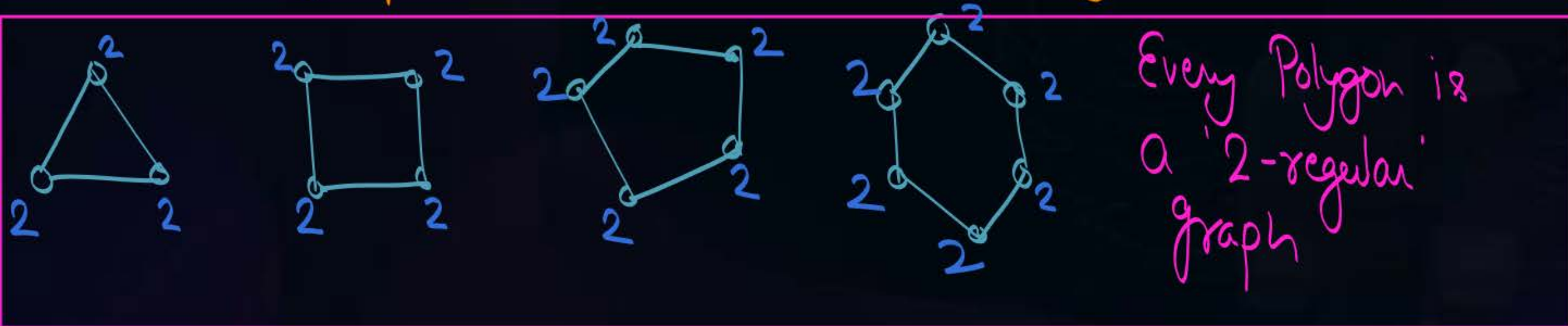
Empty Graph : - A graph with no vertex and no edges
is called an Empty graph



Topic : Regular Graphs

→ In a graph G , if degree of each vertex is Exactly same, then graph G is called a regular graph

① In a graph G , if degree of each vertex is exactly ' k ' then graph G is called a ' k -regular' graph





2 mins Summary



✓
Topic

Representation of graph

✓
Topic

Simple graph, multi graph and pseudo graph

✓
Topic

Maximum number of edges in a simple graph

✓
Topic

Maximum number of simple graph with n vertices

✓
Topic

Terminologies related to graphs

THANK - YOU